IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-SE-023

1. Topic (12 words max)

AI-Powered Research Gap Identifier & Trend Analyzer

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature and Gravity of the Problem.

Identifying genuinely novel research topics and managing supervised projects remains a persistent challenge in higher education. Undergraduate and postgraduate students often struggle to differentiate between truly novel ideas and those that closely overlap with existing studies, leading to unintentional duplication and inefficient use of academic resources. At the same time, academic supervisors are required to oversee multiple parallel projects, monitor progress, and ensure academic integrity using fragmented and largely manual processes. Research institutions further face difficulties in maintaining transparent, reliable records of idea ownership, project evolution, and contribution histories across departments and research groups.

This problem affects three key stakeholders: Student Researchers who require early validation of novelty, Academic Supervisors responsible for quality assurance and guidance, and Institutions that must maintain research standards and institutional reputation.

Impact of the Problem.

The absence of a centralized research governance framework has multi-dimensional consequences. Ethically and socially, weak mechanisms for recording idea origination and contribution create ambiguity in ownership, discouraging openness and collaboration. Research evaluation studies emphasize that academic value is strongly linked to demonstrable originality and contribution, yet these qualities are difficult to verify without reliable provenance records [1]. Technically, novelty assessment still depends heavily on manual literature reviews, which do not scale to the rapidly growing volume of scholarly publications. Advances in natural language processing, particularly deep bidirectional transformer models such as BERT, have demonstrated strong performance in semantic similarity detection and document understanding, indicating their suitability for automated

research idea comparison and validation [2]. Economically, duplicated research waste limited research funding that could be better directed toward unexplored areas. Legally, unresolved disputes over attribution can lead to significant reputational damage for individuals and universities [3]. Blockchain-based systems have further been shown to offer practical mechanisms for immutable data provenance and accountability, supporting transparent research records and academic integrity [4].

FoC Research Cluster and SDG.

This project aligns with the Centre of Excellence for AI (CoEAI) research cluster, utilizing Natural Language Processing (NLP) for similarity analysis and Recommendation Engines for collaborator matching. It targets the Education Technology (EdTech) and Research Management industry verticals. This research addresses SDG 4 (Quality Education) by enhancing mentorship, SDG 9 (Industry, Innovation, and Infrastructure) by fostering a structured innovation ecosystem, and SDG 16 (Peace, Justice, and Strong Institutions) by promoting transparency and accountability in academic research.

References.

[1] L. Bornmann and H. D. Daniel, "Are there better indices for evaluation purposes than the h-index? A comparison of nine different variants of the h-index using data from biomedicine," *J. Amer. Soc. Inf. Sci. Technol.*, vol. 59, no. 5, pp. 830–837, Mar. 2008.

[2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. 2019 Conf. North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, Minneapolis, MN, USA, 2019, pp. 4171–4186.

[3] OECD, *Scientific Advice for Policy Making: The Role and Responsibility of Expert Bodies and Individual Scientists*, OECD Science, Technology and Industry Policy Papers, no. 21. Paris, France: OECD Publishing, 2015.

[4] G. Zyskind and O. Nathan, "Decentralizing privacy: Using blockchain to protect personal data," in *Proc. IEEE Security and Privacy Workshops (SPW)*, San Jose, CA, USA, 2015, pp. 180–184.

6. Brief description of Existing Research and Systems (300 words max)

Existing Products and Systems.

Current research support technologies can broadly be grouped into discovery tools and management systems. Academic search engines and AI-assisted discovery platforms, such as Google Scholar, Semantic Scholar, Elicit, IEEE Xplore, and Bohrium, enable researchers to search large collections of scholarly publications, identify relevant studies, and obtain basic summaries. These platforms are widely used during the early stages of research for literature exploration and citation tracking.

In addition, specialized tools such as Connected Papers and Research Rabbit focus on visualizing citation networks and thematic relationships between papers, helping users understand how research topics evolve and how studies are interconnected.

On the management side, research activities are commonly coordinated using general-purpose collaboration and productivity tools such as GitHub for version control, Jira and Trello for task tracking, and institutional Learning Management Systems (LMS) for basic communication between students and supervisors. While effective individually, these tools operate independently. Discovery platforms typically stop at literature retrieval, while management systems do not consider research originality, conceptual overlap, or academic integrity.

Existing Research.

Academic research has introduced several methods for automatically analyzing scholarly content. Approaches in Natural Language Processing (NLP) has advanced from early topic modeling techniques, such as Latent Dirichlet Allocation (LDA), toward deep learning-based transformer architectures, including models such as BERT and SciBERT, which are capable of semantic similarity detection. Other studies have explored recommendation techniques to support collaborator matching and expert identification. Separately, blockchain-based approaches have been proposed to maintain trustworthy records of research data and contributions.

Gaps and Limitations.

Despite these advances, several limitations remain unresolved. First, most existing systems operate exclusively on published literature, offering little support for assessing the originality of research ideas at the proposal or early development stage. Second, current platforms do not integrate similarity analysis with accountability or reputation mechanisms that could discourage deliberate duplication of ideas. Finally, research prototypes addressing content analysis, recommendation, or data provenance are rarely embedded within a structured supervisory workflow that includes task coordination, progress tracking, and formal documentation.

As a result, no unified platform currently supports the full research lifecycle from idea formulation to supervised execution and long-term ownership management.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Addressing the Identified Gaps.

The proposed platform introduces a unified approach to research idea management by treating early-stage research concepts as formal, trackable digital assets. Unlike existing systems that focus either on literature discovery or project coordination in isolation, this solution integrates originality assessment, supervision workflows, collaboration support, and ownership tracking within a single environment.

To address pre-publication blindness, the platform evaluates research ideas at the proposal stage using semantic analysis, allowing conceptual overlap to be detected before projects begin. To overcome disconnected governance, similarity results are directly linked to a reputation and accountability mechanism that discourages intentional reuse of ideas. Finally, the system resolves lifecycle fragmentation by embedding analytical and integrity mechanisms into a structured workflow that supports continuous supervision, task allocation, and progress management. Together, these features transform research management from a collection of disconnected tools into a coherent end-to-end process.

System Perspective.

The solution follows a modular, layered architecture to ensure high performance, improve scalability, and maintain parallel development.

1. Input Layer.

The system accepts:

- Research idea submissions, including abstracts and keywords provided by students.
- User profiles containing academic background, skills, departmental affiliation, and availability.
- Metadata from external scholarly repositories to support contextual analysis and benchmarking.

2. Process Layer (Core Components).

- **Intelligence Engine:** Performs semantic embedding and similarity analysis to compare submitted ideas against existing research and ongoing internal projects. It generates quantitative overlap measures and identifies potential novelty gaps.
- **Collaboration Engine:** Constructs a dynamic academic network by modeling relationships between researchers, topics, and departments. This enables automated recommendations for suitable supervisors and collaborators.
- **Operations Workspace Module:** Provides a structured environment for research execution, including digital logbooks, task scheduling, milestone tracking, and supervisor feedback. This module standardizes supervision practices and ensures consistent documentation.
- **Integrity and Governance Module:** Records critical project events and ownership claims using cryptographic hashing on a decentralized ledger. It also maintains a reputation profile that reflects compliance with originality and ethical research practices.

3. Output Layer.

The system delivers:

- A quantified novelty and overlap assessment for each research idea.
- Ranked recommendations for supervisors and collaborators.
- A real-time progress and compliance dashboard for students and supervisors.
- A verifiable digital certificate representing research idea ownership and project history.

Conceptual Diagram.

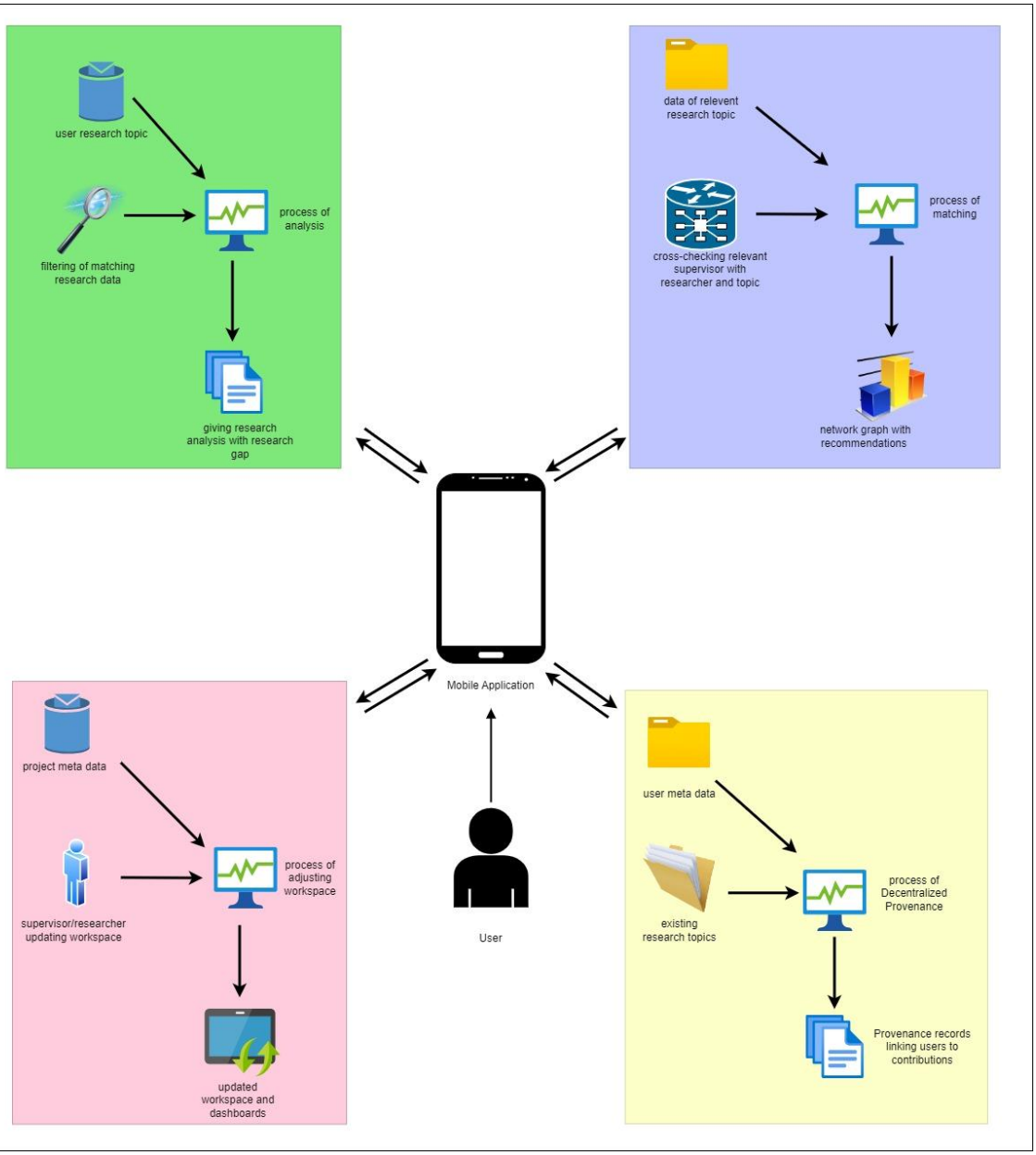


Diagram URL: https://mysliit-my.sharepoint.com/:f:/g/personal/it22629562_my_sliit_lk/IgBGC5b8njkORb63S16jPeOUAe6WScNDSErml1ULVKo1c8M?e=JZdWbH

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Availability of Specialized Domain Expertise.

Specialized knowledge of academic research workflows, supervision practices, and research integrity requirements will be obtained through interviews and surveys with relevant stakeholders (academic supervisors and lecturers) and institutional frameworks. Members of the project team will conduct consultations with stakeholders to gain first-hand insight about the activities and processes related to research projects, supervision processes, progress monitoring practices, and common challenges encountered in student research management.

Simultaneously, institutional research policies, data management guidelines, and undergraduate supervision guidelines will be reviewed in order to fully understand formal governance structures, documentation standards, and compliance requirements. This dual approach ensures that both practical operational knowledge and formal regulatory expectations are included in the system design, allowing the platform to align closely with real-world academic environments.

Technical Knowledge and Skill Development.

The project requires the team to acquire specialized technical knowledge through structured learning and guided experimentation across four core technical domains.

- **Intelligence (NLP and Similarity Analysis):** Skills in transformer-based language models and vector representations will be developed using frameworks such as Hugging Face Transformers and PyTorch**.**
- **Social/Graph (Collaboration and Recommendation):** Knowledge of graph querying (Cypher) and relationship mapping will be gained via Neo4j documentations and graph theory coursework.
- **Collaborative Workspace (Research Operations):** Skills in workflow orchestration and real-time collaboration systems will be developed using RESTful backend services, role-based access control, and event-driven architectures to support digital logbooks, milestone tracking, and supervisor feedback loops.
- **Integrity (Blockchain and Governance):** Expertise in decentralized application development will be cultivated using Solidity smart contracts within the Hardhat development environment.

Data Collection Strategies.

1. **Public Scholarly Data:** Open-access sources such as arXiv, the Semantic Scholar API, and the CORE dataset will be used to build a reference corpus for training and validating similarity detection models.
2. **User-Generated Data:** During pilot deployment, student volunteers will provide research abstracts, project descriptions, and progress logs under informed consent for system evaluation and refinement.
3. **Synthetic Data:** Artificial datasets will be generated to simulate plagiarism cases, conflicting ownership claims, and abnormal reputation score behavior to test the robustness of the governance and integrity modules.

Ethical Clearance and Data Privacy

Ethical clearance is mandatory for this project as it involves the collection of intellectual property (research ideas) and personal profiles. Following safeguards will be followed in order to achieve it.

- **Informed Consent:** Users must opt-in to before their research abstracts or profiles are processed.
- **Privacy-Preserving Provenance:** Only irreversible SHA-256 hashes of ideas and milestones will be stored on the blockchain, ensuring that original content remains off-chain and confidential.
- **Access Control:** Role-based policies will separate public metadata (research titles) from restricted content (full documents and internal evaluations), supported by authentication and audit logging mechanisms.

9. Objectives and Novelty

Main Objective

To design and evaluate an AI-driven collaborative research platform that enables early-stage research idea analysis, semantic analysis and gap detection, intelligent recommendation of collaborators and supervisors, structured research project management, and tamper-resistant governance of research idea ownership within university research environments.

Student ID	Sub Objective	Tasks	Novelty
Bandara S. M. D. M IT22141842	Semantic Analysis & Gap Detection	**1.** Collect and preprocess academic metadata, including abstracts, keywords, and citation information, from open-access scholarly sources. **2.** Design and generate semantic representations of research ideas and publications using transformer-based embedding models. **3.** Implement similarity detection mechanisms to compare Ideas and Literature **4.** Apply clustering and topic modeling to identify, emerging, or saturated emerging research areas. **5.** Evaluate system performance and accuracy using benchmark datasets and structured validation from domain experts.	**1.** Unlike existing platforms such as Elicit or Semantic Scholar, which primarily operate on published content, this component directly analyzes user-submitted research ideas. **2.** It introduces semantic novelty assessment at the ideation stage, enabling originality evaluation prior to formal project approval or publication. **3.** The integration of gap detection with idea-level similarity tracking provides early warnings against unintentional duplication, which is an ability not currently available in mainstream academic discovery or search tools.

Widanagama K. L IT22629562	Recommendation & Collaboration Network	**1.** Collect and structure researcher/supervisor profiles and project/idea metadata. **2.** Generate profile representation for supervisors, researchers, and projects. **3.** Implement matching between research idea supervisor and researcher **4.** Build a collaboration network and apply graph ranking to improve recommendations.	**1.** Existing platforms mostly recommend papers or show citation-related suggestions but this component provides idea-first matching to supervisors, collaborators, and projects before publication. **2.** Combines semantic similarity with a collaboration network graph to produce more reliable and context aware recommendations than keyword only approaches. **3.** Produces explainable recommendations which are rarely offered in current academic discovery tools.
Wijesinghe L. P. P IT22555380	Collaborative Workspace & Progress Tracking	**1.** Model the workflow of university research projects and identify key entities: projects, roles (student/supervisor), milestones, tasks, submissions, logbooks, and rubric templates. **2.** Develop the in-platform project workspace: project page, timeline/ milestones view, file/link submissions, and organized logbook/weekly update forms. **3.** Create and implement task assignment functionalities specifically designed for research modules: one-time versus recurring tasks, deadlines, reminders/	**1.** Offers a workspace that is aware of research modules (logbooks, milestones, rubrics, supervisor reviews) instead of the generic task boards typically found in Jira/Trello or basic LMS submissions. **2.** Facilitates the transition from idea discovery to supervised execution by integrating research context (topic, related papers/gaps, rationale) directly into the workflow of the workspace, rather than maintaining it separately in external literature tools.

		notifications, and supervisor review/ approval processes. **4.** Construct tools for supervisor oversight: a dashboard displaying all supervised projects, progress summaries, pending reviews, and risk indicators for missed milestones or inactivity. **5.** Establish integration connectors and evidence linking: linking GitHub repositories (commits/issues/PR activity), along with API-based linking options for Jira/Trello/Moodle/Drive where applicable.	**3.** Presents evidence-based supervision dashboards that merge structured research updates (logbooks/ submissions) with development artifacts (e.g., GitHub activity) to enhance transparency and enable early risk detection. **4.** Generates audit-friendly and privacy-controlled project records that can be directly linked to provenance/ reputation mechanisms (e.g., traceable submissions and approvals) without the need to publicly expose private documents.
Darshan R IT22097156	Integrity & Governance	**1.** Identify and model academic research events: topic submissions, revisions, supervisor/co-supervisor approvals, TAF and proposal milestones. **2.** Design a blockchain-inspired append-only ledger with hash-linked blocks and timestamping. **3.** Implement ledger storage with tamper-evident verification and explicit ownership/contribution mapping. **4.** Integrate the ledger with the collaborative research workflow system. **5.** Simulate realistic academic workflows with dummy data to validate traceability, integrity, and optional AI-assisted anomaly detection.	**1.** Applies blockchain-inspired immutability specifically to academic research governance, beyond basic versioning or task tracking. **2.** Provides end-to-end provenance, accountability, and traceability across the full research lifecycle. **3.** Ensures tamper-evident recording of topic evolution, approvals, and contributions, which existing academic tools do not formally guarantee. **4.** Optionally augmented with AI-assisted integrity analysis to flag unusual research activity patterns, enhancing auditability and research integrity.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	X	No	

b) The proposed topic exhibits novelty.

Yes	X	No	

c) The student group can successfully execute the proposed project.

Yes	X	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	X	No	

e) Any other comments:

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	Title	First Name	Last Name	Signature
Supervisor	Mr.	Jeewaka	Perera	
Co-Supervisor	Ms.	Poojani	Gunathilake	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				